

PENANG INTL, Malaysia

WMO: 486010

Lat: 5.2956N

Lon: 100.2680E

Elev: 11

StdP: 14.69

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	73.8	74.8	64.7	91.5	82.0	66.5	97.4	81.8	14.5	82.9	12.7	84.2	3.3	350	0.356

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	11.1	91.8	79.3	91.5	79.3	90.7	79.3	82.4	88.9	81.8	88.2	81.2	87.6	8.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.7	159.5	87.3	80.0	155.7	86.4	79.2	151.5	85.5	46.5	88.7	45.5	89.1	45.0	88.6	88.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.8	12.1	11.2	70.7	94.7	3.3	2.9	68.4	96.8	66.5	98.4	64.7	100.0	62.3	102.1		
(5)			66.8	83.8	1.8	1.6	65.5	84.9	64.4	85.8	63.4	86.7	62.1	87.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.8	82.7	83.3	83.7	83.6	83.6	83.4	82.9	82.4	82.0	81.8	81.9	82.1	
	DBStd	1.85	1.57	1.69	1.92	1.67	1.71	1.77	1.80	1.82	1.75	1.62	1.46	1.65	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD50	11966	1014	934	1044	1009	1042	1003	1019	1004	960	985	956	996	
	CDD65	6491	549	514	579	559	577	553	554	539	510	520	506	531	
	CDH74	72028	5932	5870	6794	6604	6885	6511	6334	5920	5374	5263	5068	5473	
CDH80	26437	2120	2323	2787	2658	2748	2574	2369	2093	1777	1648	1584	1756		
Wind	WSAvg	4.3	5.4	5.3	4.5	4.1	3.8	3.9	4.1	4.1	3.8	3.8	4.0	5.2	
Precipitation	PrecAvg	93.9	3.0	3.2	5.9	8.1	8.5	6.7	7.6	9.8	13.1	14.1	9.3	4.7	
	PrecMax	128.1	14.6	9.8	15.8	15.3	18.4	15.6	14.2	23.3	24.8	34.2	20.4	15.0	
	PrecMin	59.7	0.0	0.0	0.1	0.7	2.5	0.9	1.2	4.1	5.0	3.9	1.8	0.2	
	PrecStd	13.7	2.6	2.4	3.5	3.4	4.0	3.3	3.1	4.1	4.6	5.6	4.4	3.0	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.7	93.2	93.6	92.9	91.8	91.8	91.6	91.4	90.6	91.0	90.0	91.1	
		MCWB	76.6	77.9	78.4	81.1	80.7	80.5	79.9	79.7	79.6	79.6	79.1	78.1	
	2%	DB	90.0	91.6	91.8	91.4	91.2	91.0	90.0	89.9	89.5	89.4	89.2	89.7	
		MCWB	77.0	77.7	79.0	80.7	80.6	80.4	79.7	79.4	79.4	79.5	79.2	77.8	
	5%	DB	89.5	90.2	91.1	89.9	89.9	89.7	89.3	89.1	88.1	88.0	88.0	88.2	
		MCWB	77.2	77.7	79.1	80.3	80.4	80.0	79.6	79.4	79.3	79.4	79.3	78.0	
	10%	DB	88.0	89.4	89.6	89.2	89.2	88.8	88.0	87.8	87.4	86.9	87.0	87.5	
		MCWB	77.2	77.7	79.1	80.2	80.3	79.8	79.5	79.4	79.1	79.0	79.0	77.9	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.8	81.7	83.0	83.0	82.9	82.8	81.9	81.7	81.5	81.6	81.6	81.1
			MCDB	87.6	89.4	90.0	90.1	89.2	89.0	88.3	88.4	88.3	88.2	87.4	87.7
2%		WB	79.8	80.4	81.5	82.0	82.0	81.8	81.1	80.8	80.6	80.7	80.6	80.0	
		MCDB	86.6	87.8	88.6	88.8	88.1	87.9	87.4	87.3	86.9	87.0	86.9	86.6	
5%		WB	79.1	79.7	80.6	81.3	81.4	81.1	80.5	80.1	79.9	79.9	79.9	79.2	
		MCDB	85.7	86.9	87.7	87.8	87.4	87.1	86.7	86.3	85.9	85.9	85.9	85.4	
10%		WB	78.4	79.1	79.9	80.7	80.8	80.5	79.9	79.6	79.4	79.3	79.2	78.7	
		MCDB	85.0	86.0	86.7	86.9	86.8	86.5	85.9	85.6	85.0	84.9	84.9	84.7	
Mean Daily Temperature Range	5% DB	MDBR	12.0	12.8	12.4	11.5	11.1	11.1	11.1	11.0	11.0	10.9	11.1	11.1	
		MCDBR	13.0	13.7	13.7	12.4	12.0	12.0	11.9	12.1	12.0	12.1	12.2	12.3	
		MCWBR	6.3	6.2	5.8	5.4	5.0	5.1	5.1	5.2	5.2	5.3	5.4	5.8	
	5% WB	MCDBR	12.0	12.6	12.4	11.6	11.1	11.1	11.2	11.3	11.3	11.4	11.5	11.5	
		MCWBR	6.0	6.1	5.8	5.5	5.1	5.1	5.2	5.3	5.3	5.4	5.5	5.8	
Clear-Sky Solar Irradiance	taub	0.445	0.479	0.489	0.493	0.470	0.486	0.487	0.489	0.482	0.485	0.455	0.437		
	taud	2.370	2.242	2.245	2.238	2.322	2.290	2.271	2.254	2.275	2.265	2.357	2.410		
	Ebn at Noon	278	271	267	262	262	254	255	259	266	266	273	278		
	Edh at Noon	40	46	46	46	41	42	43	44	44	44	40	38		
All-Sky Solar Radiation	RadAvg	1618	1844	1824	1765	1622	1603	1568	1571	1509	1488	1474	1464		
	RadStd	115	109	138	65	77	72	69	77	93	125	78	118		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.59	+0.25	N/A	+0.72	+0.68	+0.49	N/A	N/A	+218
(47) Regional (10 neighbors)		+0.54	+0.32	N/A	+0.81	+0.65	+0.61	N/A	N/A	+218

Nomenclature: See separate page